

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', None)

print("Libraries loaded successfully")
```

Libraries loaded successfully

```
from google.colab import files

uploaded = files.upload()
```

traffic.csv
traffic.csv(text/csv) - 30630222 bytes, last modified: 26/5/2026 - 100% done
 Saving traffic.csv to traffic.csv

```
df = pd.read_csv("traffic.csv", encoding='latin1')

df.head()
```

	event	date	country	city	artist	album	track	isrc	
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	1c
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	1c

```
print("Shape:", df.shape)

print("\nData types:")
print(df.dtypes)

print("\nMissing values:")
print(df.isnull().sum())
```

Shape: (226278, 9)

Data types:

event	object
date	object
country	object
city	object
artist	object
album	object
track	object

```
isrc      object
linkid     object
dtype: object
```

Missing values:

```
event      0
date       0
country    11
city       11
artist     37
album      5
track      5
isrc       7121
linkid     0
dtype: int64
```

```
# Data cleaning
```

```
# Convert date column
```

```
df["date"] = pd.to_datetime(df["date"])
```

```
# Fill missing values
```

```
df["country"] = df["country"].fillna("Unknown")
```

```
df["city"] = df["city"].fillna("Unknown")
```

```
df["artist"] = df["artist"].fillna("Unknown")
```

```
df["album"] = df["album"].fillna("Unknown")
```

```
df["track"] = df["track"].fillna("Unknown")
```

```
df["isrc"] = df["isrc"].fillna("Unknown")
```

```
# Check duplicates
```

```
print("Duplicate rows:", df.duplicated().sum())
```

```
print("\nMissing values after cleaning:")
```

```
print(df.isnull().sum())
```

```
Duplicate rows: 103711
```

```
Missing values after cleaning:
```

```
event      0
```

```
date       0
```

```
country    0
```

```
city       0
```

```
artist     0
```

```
album      0
```

```
track      0
```

```
isrc       0
```

```
linkid     0
```

```
dtype: int64
```

```
# Remove duplicate rows
```

```
df = df.drop_duplicates()
```

```
print("New dataset shape:", df.shape)
```

```
print("Remaining duplicates:", df.duplicated().sum())
```

New dataset shape: (122567, 9)
Remaining duplicates: 0

```
# Website traffic metrics

total_events = len(df)

unique_users = df["linkid"].nunique()

top_events = df["event"].value_counts()

print("Total Events:", total_events)
print("Unique Users/Sessions:", unique_users)

print("\nEvent Distribution:")
print(top_events)
```

Total Events: 122567
Unique Users/Sessions: 3839

Event Distribution:

event	
pageview	73360
click	32499
preview	16708

Name: count, dtype: int64

```
# Event distribution chart

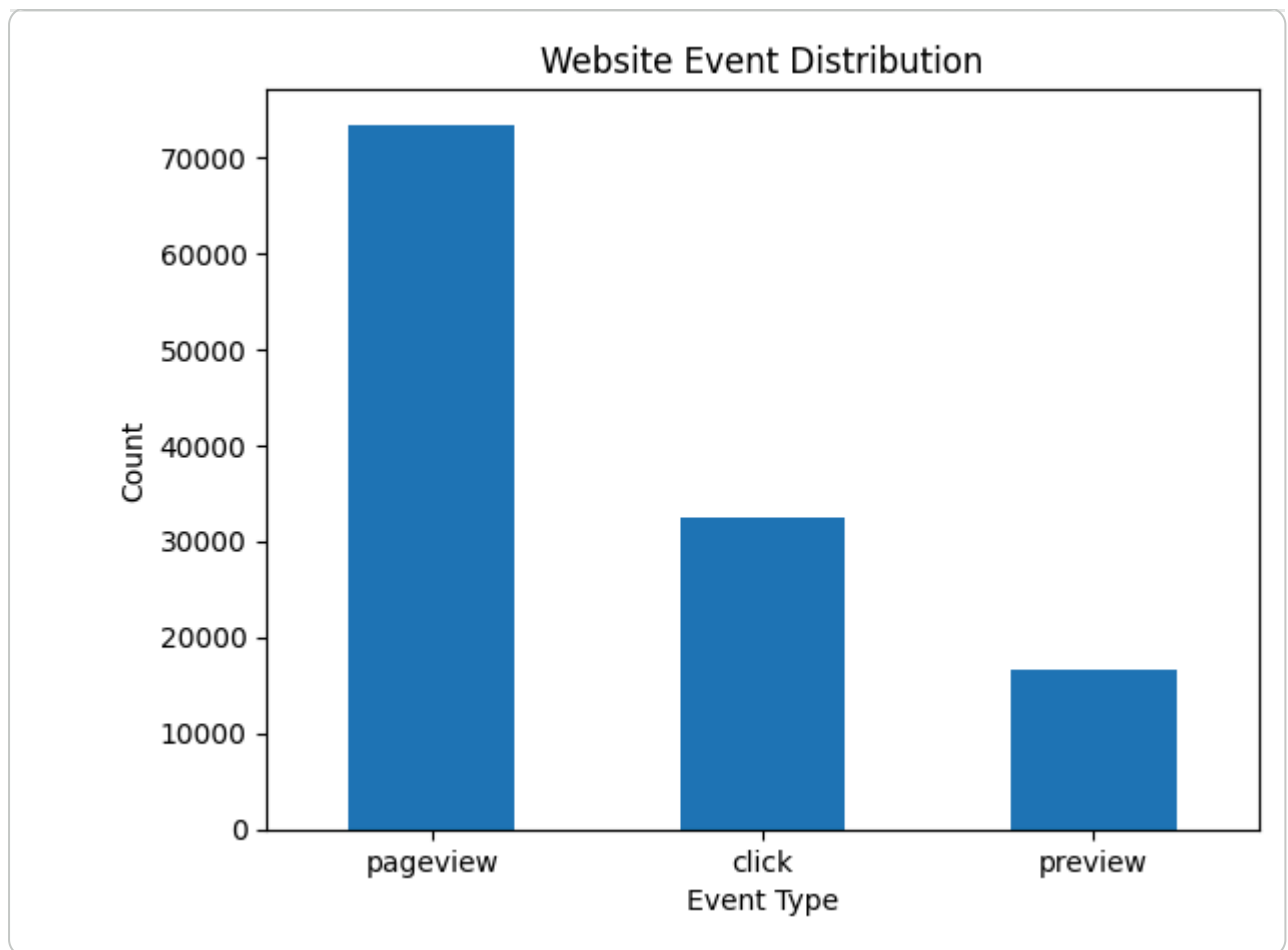
event_counts = df["event"].value_counts()

event_counts.plot(kind="bar")

plt.title("Website Event Distribution")
plt.xlabel("Event Type")
plt.ylabel("Count")

plt.xticks(rotation=0)

plt.show()
```



Observation: Website Event Distribution

- Pageviews recorded the highest activity with 73,360 events, indicating that users frequently visit website pages.
- Click events accounted for 32,499 interactions, showing that a portion of visitors actively engage with content.
- Preview events were the lowest at 16,708, suggesting fewer users proceed deeper into content interaction.
- There is a noticeable drop from pageviews to clicks, indicating a potential conversion gap.
- User engagement decreases as users move through the interaction journey.

Business Insight:

The website attracts strong initial traffic, but user progression through the engagement funnel decreases. Improving call-to-action visibility, page design, and content relevance could help convert more visitors into active users.

```
# Top 10 countries by traffic

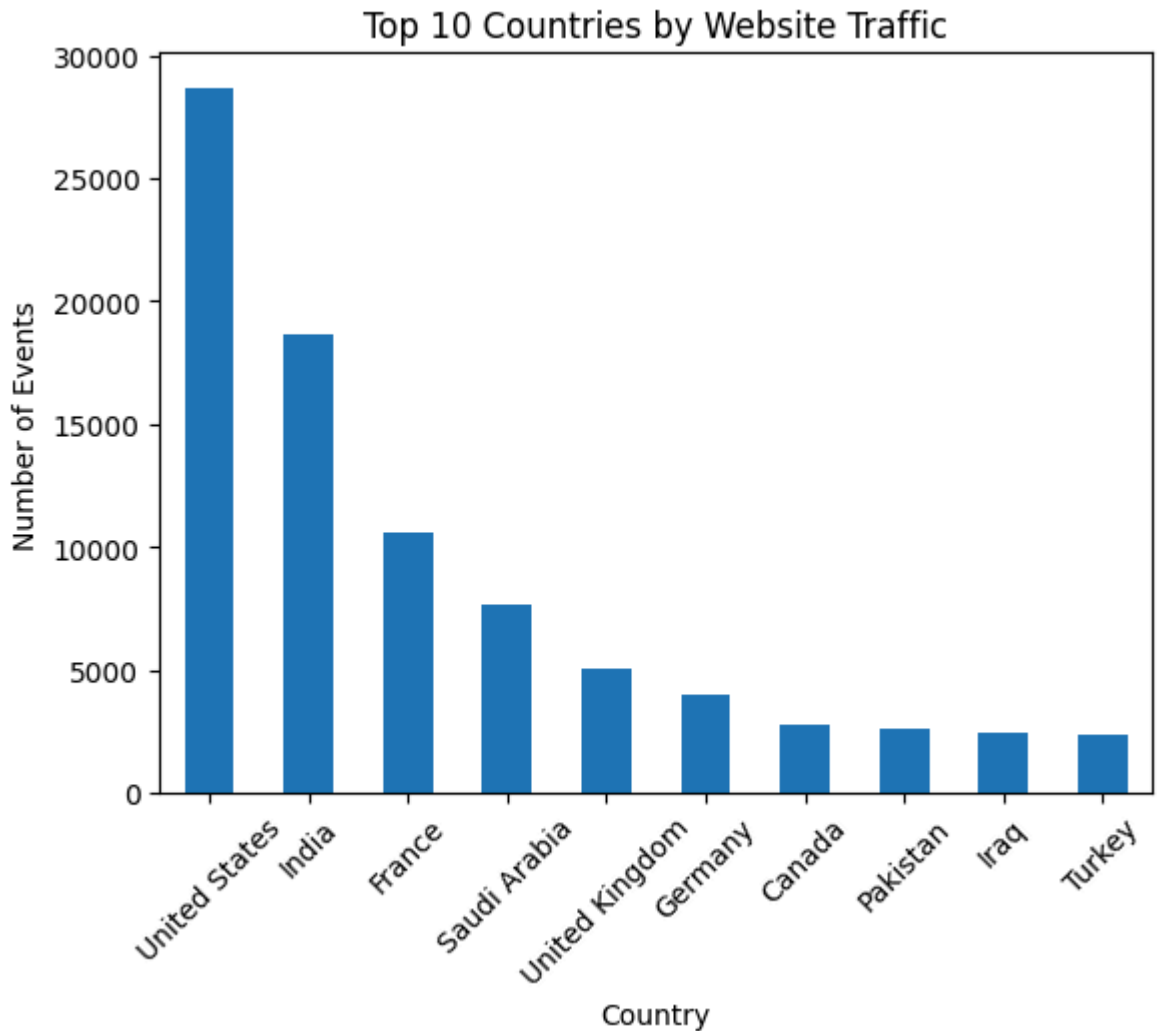
top_countries = df["country"].value_counts().head(10)

top_countries.plot(kind="bar")
```

```
plt.title("Top 10 Countries by Website Traffic")
plt.xlabel("Country")
plt.ylabel("Number of Events")

plt.xticks(rotation=45)

plt.show()
```



Observation: Top Countries by Website Traffic

- The United States generated the highest website traffic with approximately 29,000+ events, making it the strongest traffic source.
- India recorded the second-highest traffic with around 19,000 events, showing significant user engagement.
- France and Saudi Arabia contributed moderate traffic levels.
- Remaining countries such as United Kingdom, Germany, Canada, Pakistan, Iraq, and Turkey generated comparatively lower traffic.
- Website traffic distribution is uneven across countries, with a few regions contributing most of the activity.

Business Insight:

The website has strong market reach in the United States and India. Alfido Tech can increase growth by strengthening campaigns in high-performing regions while developing targeted marketing strategies in lower-performing countries.

```
# Top 10 artists by traffic

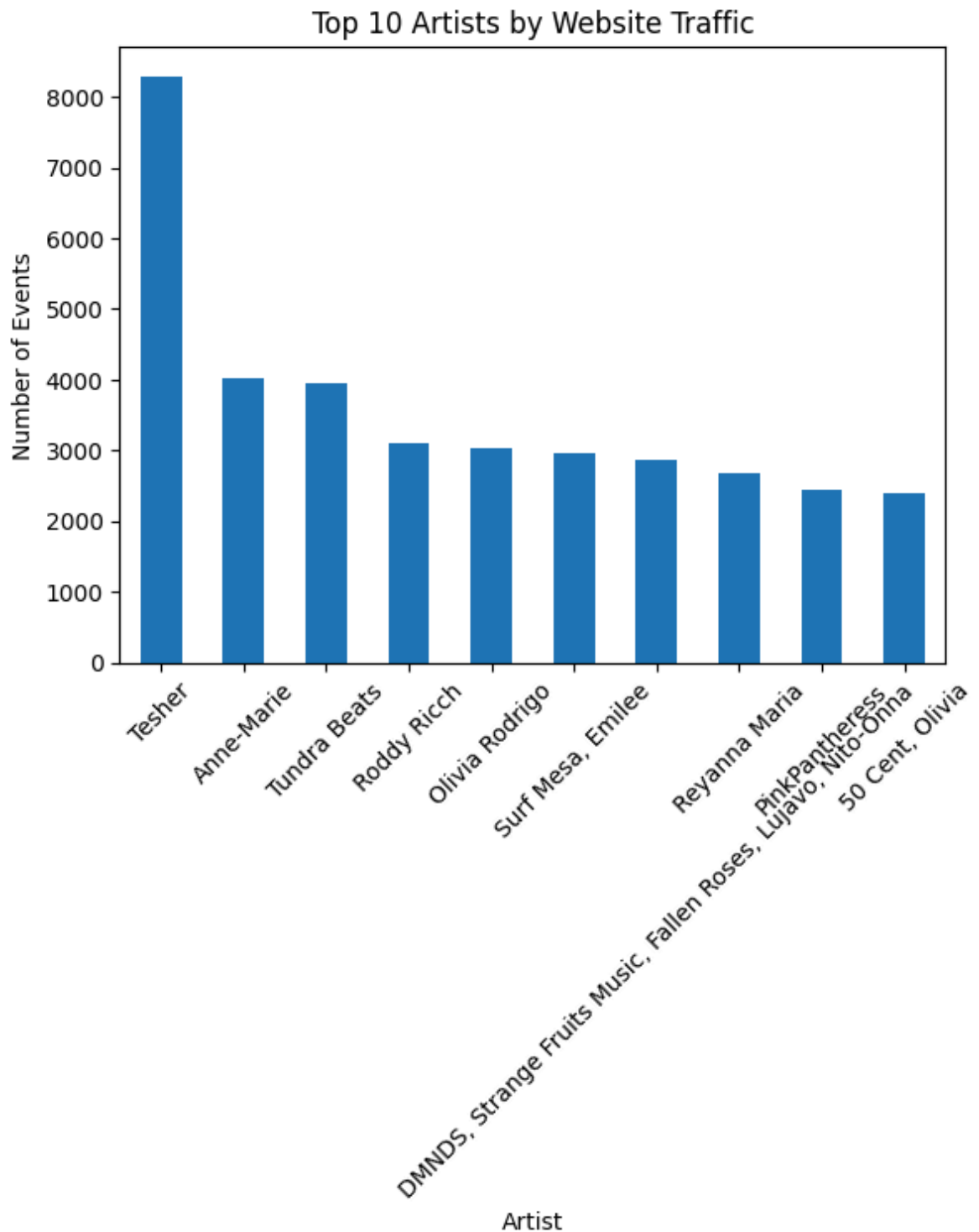
top_artists = df["artist"].value_counts().head(10)

top_artists.plot(kind="bar")

plt.title("Top 10 Artists by Website Traffic")
plt.xlabel("Artist")
plt.ylabel("Number of Events")

plt.xticks(rotation=45)

plt.show()
```



Observation: Top Artists by Website Traffic

- Teshar generated the highest website traffic with approximately 8,000+ events, significantly outperforming other artists.
- Anne-Marie and Tundra Beats showed strong engagement and ranked among the top-performing content creators.
- Traffic gradually declines after the top few artists, indicating that user attention is concentrated on limited content.
- The gap between the top performer and lower-ranked artists suggests unequal content engagement.

- A small number of artists contribute a large proportion of user interactions.

Business Insight:

Popular content drives a major portion of website engagement. Alfido Tech can improve user retention by promoting high-performing content, recommending related content, and improving visibility for lower-performing content.

Actionable Recommendations for Alfido Tech**1. Improve click-through conversion**

A large gap exists between pageviews and click events. Improve call-to-action buttons, page layouts, and content placement to encourage users to interact more frequently.

2. Focus on high-traffic regions

The United States and India generated the highest website activity. Increase targeted campaigns and marketing investments in these regions to maximize returns.

3. Promote related content

A small number of artists generate most traffic. Recommend similar content to users to increase engagement across less popular content.

4. Optimize lower-performing markets

Countries with lower traffic should receive localized promotions and region-specific campaigns to improve user acquisition.